Intro to Machine Learning
Practice Problems 2: Ridge Regression, the Lasso and the Elastic Net

Practice problems, not turned in — Solutions

Directions: These practice problems go with Week 2 (ridge regression, the lasso and the elastic net). They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Each problem is adapted from the exercise cited next to it; see the book for the original wording.

Throughout, n is the number of observations, p the number of predictors, and the intercept β_0 is never penalized. Unless a problem says otherwise, ridge and the lasso use ISL's scaling:

$$\text{ridge: } \mathrm{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2, \qquad \text{lasso: } \mathrm{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|, \qquad \mathrm{RSS} = \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j \Big)^2 .$$

1. **More or less flexible than least squares?** (ISL 6.6, Exercise 2(a)–(b); ISL 6.2.) For each method below, choose the one correct statement and justify it.

   (a) The lasso, compared with least squares, is

   i. more flexible, so it predicts better when its increase in bias is smaller than its decrease in variance;

   ii. more flexible, so it predicts better when its increase in variance is smaller than its decrease in bias;

   iii. less flexible, so it predicts better when its increase in bias is smaller than its decrease in variance;

   iv. less flexible, so it predicts better when its increase in variance is smaller than its decrease in bias.

   (b) The same question for ridge regression compared with least squares.

   Solution.

   (a) **iii.** The lasso minimizes RSS over a restricted set, $\sum_j |\beta_j| \le s$, so it can represent fewer fits than least squares: it is *less* flexible. Shrinking the coefficients toward zero (and setting some exactly to zero) adds bias, because the estimates are no longer centered on the true β. In exchange the fit depends less on the particular training sample, so its variance falls. Test error is $\sigma^2 + \text{bias}^2 + \text{variance}$, so the lasso wins exactly when the variance it removes is larger than the bias^2 it adds.

   (b) **iii**, for the same reason. Ridge restricts $\sum_j \beta_j^2 \le s$, so it is less flexible than least squares. It trades a little bias for lower variance, and it predicts better when the drop in variance

outweighs the added bias^2. A classic result (Hoerl and Kennard, 1970) shows that for ridge such a $\lambda > 0$ always exists: the bias^2 grows like λ^2 near $\lambda = 0$ while the variance falls like λ, so the first bit of shrinkage is free. (Options i and ii are wrong because neither method is more flexible than least squares; option iv has the trade backwards.)

2. **The lasso as the budget grows.** (ISL 6.6, Exercise 3; ISL 6.2.2.) Fit the lasso in its constrained form,

$$\text{minimize RSS} \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \le s,$$

and increase s from 0. For each quantity below, say whether it (i) increases and then decreases (an inverted U), (ii) decreases and then increases (a U), (iii) steadily increases, (iv) steadily decreases, or (v) stays constant. Justify each answer.

(a) training RSS (b) test RSS (c) variance (d) squared bias (e) irreducible error

Solution. At $s = 0$ every slope is zero and the fit is $\bar{y}$, the least flexible linear model. As s grows the constraint loosens, and once $s \ge \sum_j |\hat{\beta}_j^{\text{OLS}}|$ the constraint is inactive and the fit is least squares. So increasing s moves from the least to the most flexible linear fit.

(a) **(iv) Steadily decreases.** Increasing s enlarges the set we minimize RSS over, and a minimum over a larger set can't be larger. It falls until s reaches $\sum_j |\hat{\beta}_j^{\text{OLS}}|$ and is flat at the OLS value after that.

(b) **(ii) A U shape.** Test RSS starts high (at $\bar{y}$ the model ignores real signal, so the bias is large), falls as the important coefficients are allowed in, and then rises again as the model starts fitting noise. That is the usual pattern; if almost every predictor matters, the right-hand rise can be small.

(c) **(iii) Steadily increases.** At $s = 0$ the fit is $\bar{y}$, whose variance is only σ^2/n. A larger budget lets the coefficients respond more to the particular sample, so the variance grows, up to the OLS variance $(p+1)\sigma^2/n$ per observation.

(d) **(iv) Steadily decreases.** At $s = 0$ the bias is as large as it gets; as s grows the estimates can move toward the true coefficients. Once the constraint stops binding (OLS), the bias is zero if the true model is linear in these predictors.

(e) **(v) Constant.** The irreducible error $\sigma^2 = \text{Var}(\varepsilon)$ is a property of the data-generating process, not of the fitting method.

3. **Ridge as the penalty grows.** (ISL 6.6, Exercise 4; ISL 6.2.1.) Now fit ridge regression in its penalized form, $\text{RSS} + \lambda \sum_j \beta_j^2$, and increase λ from 0. Answer the same five parts as Problem 2, using the same options i–v.

Solution. Increasing λ runs the path of Problem 2 in the *opposite* direction: $\lambda = 0$ is least squares and $\lambda \to \infty$ gives $\bar{y}$. So each answer flips, except (e).

(a) **(iii) Steadily increases.** $\lambda = 0$ gives the smallest possible training RSS (least squares). A bigger penalty pulls the coefficients further from the RSS minimizer, so training RSS rises toward $\sum_i (y_i - \bar{y})^2$.

(b) **(ii) A U shape.** Test RSS first falls, because shrinkage removes variance faster than it adds bias (Problem 1(b) explains why the first bit of shrinkage helps), and then rises once the coefficients are shrunk so much that real signal is lost.

(c) **(iv) Steadily decreases.** Using the singular value decomposition $X = UDV^T$ (ESL 3.4.1), the variance of the fitted values is $\sigma^2 \sum_j \left(d_j^2/(d_j^2+\lambda)\right)^2$, and each term decreases in λ.

(d) **(iii) Steadily increases.** The bias^2 of the fitted values is $\sum_j \left(\lambda/(d_j^2+\lambda)\right)^2 \gamma_j^2$, where $\gamma_j = u_j^T X\beta$, and each factor $\lambda/(d_j^2+\lambda)$ increases in λ.

(e) **(v) Constant**, for the same reason as Problem 2(e).